SAHUL HAMEED

Software Engineer — Backend & Data Systems (Node.js, Python, Distributed Pipelines, Kafka)
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SUMMARY

Full-stack engineer with nearly 3 years building TypeScript/Node.js microservices and distributed Python data pipelines (Kafka, HDFS) that process high-volume network telemetry at scale, alongside end-to-end product delivery. Comfortable owning a system from packet ingestion and data lake architecture to customer-facing dashboards.

PROFESSIONAL EXPERIENCE

Software Engineer — Nubewell Networks (Network Traffic Optimization) *01/2025 - Present*

Bengaluru | Project: Network Traffic Analytics & Monitoring Platform

- **Architected 6 TypeScript/Express.js microservices** (analytics, reports, config, alerts, auth, log-server) deployed on Docker; the config service manages NetFlow device onboarding via MongoDB aggregation pipelines and HTTP proxying to push configuration changes directly to live network appliances.
- **Built a structured log pipeline** — services emit JSON logs to disk; Fluent Bit tails log files, classifies records by type (system vs. accounting) using tag-rewrite rules, and publishes to two dedicated Kafka topics; a KafkaJS consumer batches ingestion at 500 records / 2s into a centralized store with a filtered Express REST API.
- **Built the analytics service** — router-filtered NetFlow record queries backed by MongoDB; session time-series API with adaptive bucket sizing that auto-selects minute/hour/day/month resolution targeting ~60 data points per query range; hourly cron-scheduled HDFS storage tracking per router; and IPDR subscriber-level flow record access.
- **Developed the reports service** — PDF generation via Puppeteer and pdfkit-table, node-cron scheduled reports (once / weekly / monthly), and singleton background services: network bandwidth sampled every 1s and aggregated per minute to MongoDB, router inactivity alerts fired after 10 min of no flow data, and HDFS storage notifications triggered at 60% capacity.
- **Engineered a multi-process Python NetFlow collector** — dedicated OS process per router sustaining ~12,000 records/sec, bypassing Python's GIL for CPU-bound UDP packet parsing, with tuned UDP socket buffers to prevent packet loss under load and dual-trigger Kafka publishing (100K records / 15s).
- **Designed the Kafka-to-HDFS ingestion layer** — 3 consumer processes enriching flows with customer IDs via a hybrid hash map (exact IPs) and binary search (IP ranges), batching into zstd-compressed Parquet targeting <=45s end-to-end latency; used MongoDB as a live-traffic layer with a rolling 1,000-flow snapshot per router for real-time dashboard queries; optimized report performance via year/month/day/hour partition pruning and vectorized integer IP range filtering; migrated report export to XLSX write-only streaming mode enabling 500,000+ row exports at constant memory.

MERN Stack Developer — Ariveguru Technology Solutions *12/2023 - 12/2024*

Bengaluru

- **Built a reusable Dynamic Form engine in React** that converts structured JSON schemas into fully rendered UI forms, eliminating manual form-building effort and improving IRB application submission speed.
- **Developed a customer-facing web portal** (React, Redux) simplifying complex IRB application workflows, with backend RESTful APIs (Express.js) and MongoDB/Mongoose for structured storage and querying of application data.
- **Built a WhatsApp query-answering POC** integrating the Meta Graph API for message handling with an LLM API for intelligent, context-aware responses to customer queries, demonstrating end-to-end conversational AI integration.

SKILLS

Languages & Backend: Python (FastAPI), Node.js (TypeScript, Express.js), JavaScript
Data & Infrastructure: Kafka, HDFS, PostgreSQL, Redis, Fluent Bit, Docker, Parquet, MongoDB
Frontend: React, Redux
AI Tools: Claude Code, Antigravity

EDUCATION

B.E., Electrical and Electronics Engineering *08/2019 - 05/2023*

Government College of Engineering, Srirangam